Question: 1 [Marks: 1]

The inverse Jacobian for polar coordinates is defined by _____

Choices:

- $\frac{\partial(x,y)}{\partial(r,\theta)}$
- $\frac{\partial(r,\theta)}{\partial(x,y)}$
- $\frac{\partial(x,r)}{\partial(y,\theta)}$ correct
- $\frac{\partial(r,y)}{\partial(x,\theta)}$

The Kronecker delta δ_{ij} = _____

Choices:

- $\begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$

correct

- $\begin{cases} 0 & i = j \\ 1 & i \neq j \end{cases}$
- $\begin{cases} 1 & i = j \\ -1 & i \neq j \end{cases}$
- $\begin{cases} -1 & i = j \\ 0 & i \neq j \end{cases}$

A tensor A^{ij} is said to be anti-symmetric if

Choices:

- $A^{ij} = -A^{ji}$ correct
- $A^{ij} = A^{ji}$
- $A^{ij} = A_{ij}$
- $A^{ij} = A_{ji}$

In the expression $y_i = c_i^r a_{rs} x_s$, ____ and ____ are dummy indices, and ____ is a free index

Jo y k sath hy woh free indices hy aur dusri side py dummy hn i k ilawa

Choices:

- r, s, i

correct

- i, r, s
- r, i, s
- s, i, r

J. W. Gibbs developed the concept of dyadic in ____

Choices:

1884

correct

1883

1885

1886

An inner product of two tensors may be characterized as a contraction of their outer product.

Choices:

True

false

true I think

Choose the expression that represents an inner product of A^{ij} and B_{mn}

Choices

- $A^{ij}B_{mn}$ correct
- $A^{ij}B_{in}$
- $A^{ji}B_{nm}$
- $A^{ii}B_{mm}$

The transformation law $\bar{x}^i = a^i_j x^j$ with $a^i_j \neq 0$ defines _____ tensors when a^i_j are orthogonal.

Choices:

- Cartesian
- Affine
- Symmetric
- Anti-symmetric

The general expression of ds^2 = _____

Choices:

$g^{ij}dx_i dx_j$

$g_{ij}dx_i dx_j$

$g^{ij}dx^i dx^j$

$g_{ij}dx^i dx^j$

correct

In general, $g_{ij} =$ _____

$e_i \cdot e_j$

Choices:

- $\vec{e}_i \cdot \vec{e}_j$
- $\hat{e}_i \cdot \hat{e}_j$
- $\vec{e}_i \cdot \vec{e}^j$
- $\vec{e}^i \cdot \vec{e}^j$

Which of the following is an absolute tensor?

Choices:

- A_{ij}
- ε^{ijk}
- δ^{ij}
- *None of these*

A _____ slides an object a fixed distance in a given direction.

Choices:

translation

rotation

scale

deformation

A _____ slides an object a fixed distance in a given direction.

Choices:

- translation
- rotation
- scale
- deformation

The general expression of arc length between $a \leq t \leq b$ is defined by

$$L = \int_a^b \sqrt{g_{ij} \frac{dx^i}{dt} \frac{dx^j}{dt}} \, dt$$

Choices:

- $L = \int_a^b \sqrt{g_{ij} \frac{dx^i}{dt} \frac{dx^j}{dt}} \, dt$
- $L = \int \sqrt{g_{ij} \frac{dx^i}{dt} \frac{dx^j}{dt}} \, dt$
- $L = \int_a^b \sqrt{\frac{dx^i}{dt} \frac{dx^j}{dt}} \, dt$
- $L = \int_a^b \sqrt{g_{ij} \frac{dx^i}{dt} \frac{dx^j}{dt}}$

The general expression of arc length between $a \leq t \leq b$ is defined by

Choices

- $L = \int_a^b \sqrt{g_{ij} \frac{dx^i}{dt} \frac{dx^j}{dt}}\, dt$
- $L = \int \sqrt{g_{ij} \frac{dx^i}{dt} \frac{dx^j}{dt}}\, dt$
- $L = \int_a^b \sqrt{\frac{dx^i}{dt} \frac{dx^j}{dt}}\, dt$
- $L = \int_a^b \sqrt{g_{ij} \frac{dx^i}{dt} \frac{dx^j}{dt}}$

Which one is the expression for $\vec{\nabla} \times \vec{A}$?

Choices:

- $\frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1\hat{e}_1 & h_2\hat{e}_2 & h_3\hat{e}_3 \\ \frac{\partial}{\partial u_1} & \frac{\partial}{\partial u_2} & \frac{\partial}{\partial u_3} \\ A_1 h_1 & A_2 h_2 & A_3 h_3 \end{vmatrix}$
- $\begin{vmatrix} h_1\hat{e}_1 & h_2\hat{e}_2 & h_3\hat{e}_3 \\ \frac{\partial}{\partial u_1} & \frac{\partial}{\partial u_2} & \frac{\partial}{\partial u_3} \\ A_1 h_1 & A_2 h_2 & A_3 h_3 \end{vmatrix}$
- $\begin{vmatrix} h_1\hat{e}_1 & h_2\hat{e}_2 & h_3\hat{e}_3 \\ \frac{\partial}{\partial u_1} & \frac{\partial}{\partial u_2} & \frac{\partial}{\partial u_3} \\ A_1 & A_2 & A_3 \end{vmatrix}$
- $\frac{1}{h_1 h_2 h_3} \begin{vmatrix} \hat{e}_1 & \hat{e}_2 & \hat{e}_3 \\ \frac{\partial}{\partial u_1} & \frac{\partial}{\partial u_2} & \frac{\partial}{\partial u_3} \\ A_1 h_1 & A_2 h_2 & A_3 h_3 \end{vmatrix}$

The inverse metric for 2-dimensional rectangular coordinates is ___

Choices:

- $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$
- $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
- $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
- $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$

Which expression is used to lower the index of A^i?

Choices

- $A_i = g^{ij}A^j$
- $A_i = g^{ij}A^i$
- $A_i = g_{ij}A^j$
- $A_i = g_{ij}A^i$

$\hat{\theta}_1 \cdot \hat{\theta}_2 \times \hat{\theta}_3 =$ _____

Choices:

1

0

-1

None of these

The transformation coefficients $a_{ij} =$ _____

Choices:

- $\cos(\hat{e}_i, \hat{e}'_j)$
- $\sin(\hat{e}_i, \hat{e}'_j)$
- $\tan(\hat{e}_i, \hat{e}'_j)$
- $\cot(\hat{e}_i, \hat{e}'_j)$

If $\vec{r} = \vec{r}(x^1, x^2, x^3)$, and $x^i = x^i(t)$ then $\frac{d\vec{r}}{dt} =$ _____

Choices:

- $\frac{\partial \vec{r}}{\partial x^1}\frac{dx^1}{dt} + \frac{\partial \vec{r}}{\partial x^2}\frac{dx^2}{dt} + \frac{\partial \vec{r}}{\partial x^3}\frac{dx^3}{dt}$
- $\frac{\partial \vec{r}}{\partial x^1}\frac{dx^1}{dt} + \frac{\partial \vec{r}}{\partial x^2}\frac{dx^2}{dt}$
- $\frac{\partial \vec{r}}{\partial x^1}\frac{dx^1}{dt}\frac{dx^2}{dt}\frac{dx^3}{dt}$
- $\frac{\partial \vec{r}}{\partial x^1}\frac{dx^1}{dt}\frac{\partial \vec{r}}{\partial x^2}\frac{dx^2}{dt}\frac{\partial \vec{r}}{\partial x^3}\frac{dx^3}{dt}$

The generalized inner product of two vectors is defined by_____

Choices

- $u^i v_i = g_{ij} u^i v^j$
- $u^i v^j = g_{ij} u^i v^j$
- $u^i v^j = g_{ij} u^i v^j$
- $u^i v^j = g^{ij} u^i v^j$

Question 27 (Marks 1)

The gradient is an example of a _____ vector.

Choices

covariant

contravariant

The gradient is an example of a _____ vector

Choices:

covariant

contravariant

The gradient is an example of a _____ vector

The gradient is an example of a _____ vector

Choices:

covariant

contravariant

Compute the arc length $L = \int_0^1 \left| \frac{d\vec{r}}{dt} \right| dt$ for $\vec{r} = t\vec{e}_1 + 2t\vec{e}_2$

Answer:

Write transformation equations from spherical to rectangular coordinates.

Answer:

Compute the formula for the tensor component $\hat{T}_{11}$ in polar coordinate system.

if the components T_{ij} are the components in rectangular coordinate system.

Answer:

Explain covariant vectors.

Answer

Which of the following represent the plane $z = 0$ in terms of spherical coordinates?

Choices:

- $\theta = \frac{\pi}{3}$
- $\varphi = \frac{\pi}{2}$
- $\theta = 0$
- $\theta = \pi$

In polar coordinates, $\frac{\partial r}{\partial x} =$ _____

Choices:

0

$\sin\theta$

$\frac{1}{\cos\theta}$

1

The index notation for gradient operator is defined as _____

Choices

- $\nabla\varphi = \frac{\partial\varphi}{\partial x_i}\hat{e}_i$
- $\nabla\varphi = \frac{\partial\varphi}{\partial x_i}$
- $\nabla\varphi = \frac{\partial\varphi}{\partial x_j}\hat{e}_i$
- $\nabla\varphi = \frac{\partial\varphi}{\partial x_i}\hat{e}_j$

The index form of Laplacian operator is _____

Choices:

$\nabla^2\varphi = \nabla\varphi$

$\nabla^2\varphi = \varphi_{,i}$

$\nabla^2\varphi = \varphi_{,ii}$

$\nabla^2\varphi = \varphi_{,ij}$

Question 23 (Marks: 1)

The index form of Laplacian operator is _____

$\nabla^2 \varphi = \varphi_{ii}$

Choices:

- $\nabla^2 \varphi = \nabla \varphi$
- $\nabla^2 \varphi = \varphi_{,i}$
- $\nabla^2 \varphi = \varphi_{,ii}$
- $\nabla^2 \varphi = \varphi_{,ij}$

[illegible]

Question 22 (Marks: 1)

Which one of the following is the scale factors for polar coordinates?

Choices:

- $h_1 = h_2 = 1$
- $h_1 = 1, h_2 = r$
- $h_1 = r, h_2 = \theta$
- $h_1 = h_2 = r$

correct

Question 21

The norm of a covariant vector is defined as _____

Choices:

- $\|v_i\| = \sqrt{g_{ij} v^i v^j}$
- $\|v_i\| = \sqrt{g^{ij} v^i v^j}$
- $\|v_i\| = \sqrt{g^{ij} v_i v_j}$ correct
- $\|v_i\| = \sqrt{g_{ij} v_i v_j}$

Question 21

The norm of a covariant vector is defined as _____

Choices:

- $\|v_i\| = \sqrt{g_{ij} v^i v^j}$
- $\|v_i\| = \sqrt{g^{ij} v^i v^j}$
- $\|v_i\| = \sqrt{g^{ij} v_i v_j}$
- $\|v_i\| = \sqrt{g_{ij} v_i v_j}$

or tension.

J. W. Gibbs in (1884) developed the concept of dyadics (object with two parts or elements) and polyadics (an object comprising of many elements) which are actually tensors in

1) m (scalar) tensor of rank zero

2) v_α . (vector) tensor of rank 1,
v^β (vector) tensor of rank 1

3) $T_{\mu\delta}$ (tensor) tensor of rank 2

4)
$A^{\mu}_{\nu\mu}$ (tensor) tensor of rank 1

2. Each index can appear at most twice in any term.

3. Each term must contain identical non-repeated indices.

So,

$$M_{ij}v_j = \sum_j M_{ij}v_j$$

is a valid expression whereas

$$M_{ij}u_jv_j + w_i \quad and \quad T_{ijk}u_k + M_{ip}$$

are not valid expressions.

Kronecker Delta

The Kronecker delta is defined as

$$\delta_{ij} = \delta_i^j = \delta^{ij} = \begin{cases} 1; & i = j \\ 0; & i \neq j \end{cases}$$

Clearly $\delta_{ij} = \delta_{ji}$ for all i, j.

The Kronecker-δ can be thought of as the identity matrix, for example for $i \equiv 1,2,3$ and $j \equiv 1,2,3$

$$\delta_{ij} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Example: The permutation symbol satisfies :

i) $\delta_{ij}\varepsilon_{ijk} = 0$

ii) $\varepsilon_{ipq}\varepsilon_{jpq} = 2\delta_{ij}$

iii) $\varepsilon_{ijk}\varepsilon_{ijk} = 6$

where δ_{ij} is the Kronecker delta.

OR $$\varepsilon_{ijk} = \frac{1}{2}(i-j)(j-k)(k-i)$$

For the indices (i, j, k) in ε_{ijk}, the values 1, 2, 3 occurring in the cyclic order (1, 2, 3) correspond to $\varepsilon = +1$, while occurring in the reverse cyclic order correspond to $\varepsilon = -1$, otherwise $\varepsilon = 0$.

- **Transpose of a Matrix**

The transpose of a matrix in index notation is defined as

$$A^T = [a_{ij}]^T_{mn} = [a^t{}_{ij}]_{nm}, \text{ where } a^t{}_{ij} = a_{ji}$$

A matrix A is orthogonal if $A^T = A^{-1}$

For example the transformation

$$(x, y) \rightarrow (x + 5, y + 3)$$

applications.

A transformation is used to change the object's

- Position (translation)
- Size (scaling)
- Orientation (rotation)
- Shapes (shear)

Hence the transformation coefficients form an orthogonal matrix and

$$\alpha_{ik}\alpha_{jk} = \delta_{ij}, \qquad \alpha_{ki}\alpha_{kj} = \delta_{ij}$$

These are called orthogonality relations.
Where orthogonal matrices satisfy the relation

$$\det\left[\alpha_{ij}\right] = \pm 1$$

After using the transformation law for coordinates of points we obtain

$$A_i = \alpha_{ij}\check{A}_j$$
$$and\ \check{A}_i = \alpha_{ji}A_j$$

Arc length in general curvilinear coordinates

The square of the element of arc length in general curv
coordinates can be expressed by

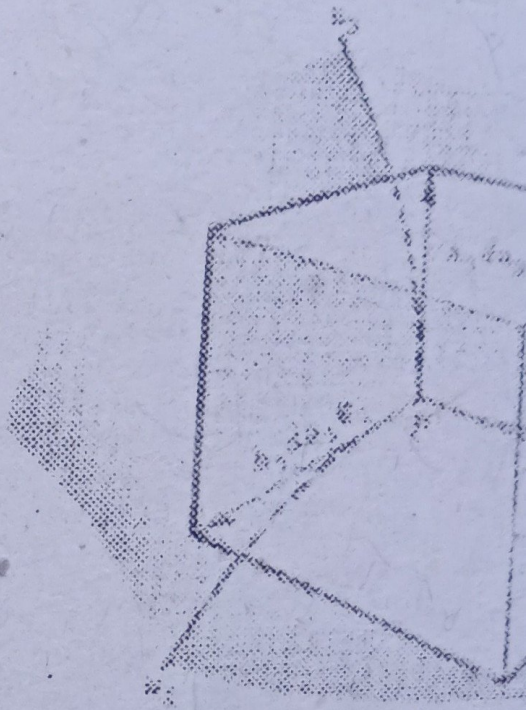
$$ds^2 = \sum_{i=1}^{3}\sum_{j=1}^{3} h_i h_j \left(\hat{e}_i . \hat{e}_j\right) du_i du_j$$

$$\bar{r} = \bar{r}(u_1, u_2, u_3)$$

$$d\bar{r} = \frac{\partial \bar{r}}{\partial u_1} du_1 + \frac{\partial \bar{r}}{\partial u_2} du_2 + \frac{\partial \bar{r}}{\partial u_3} du_3$$

$$d\bar{r} = h_1 du_1 \hat{e}_1 + h_2 du_2 \hat{e}_2 + h_3 du_3 \hat{e}_3$$

$$ds^2 = d\bar{r} . d\bar{r}$$



Arc length element in orthogonal curvilinear coordinates

The arc length ds is the length of the infinitesimal vector $d\vec{r}$:

$$(ds)^2 = d\vec{r}\cdot d\vec{r}$$

In curvilinear coordinates, we have

$$\vec{r} = \vec{r}(u_1, u_2, u_3)$$

$$d\vec{r} = \frac{\partial \vec{r}}{\partial u_1} du_1 + \frac{\partial \vec{r}}{\partial u_2} du_2 + \frac{\partial \vec{r}}{\partial u_3} du_3$$

$$d\vec{r} = h_1 du_1 \hat{e}_1 + h_2 du_2 \hat{e}_2 + h_3 du_3 \hat{e}_3$$

$$ds^2 = h_1^2 du_1^2 + h_2^2 du_2^2 + h_3^2 du_3^2$$

the space

$$\nabla = \frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k$$

- Gradient $\rightarrow \nabla \psi$
- Divergence $\rightarrow \nabla . \vec{A}$
- Curl $\longrightarrow \nabla \times A$
- Laplacian $\rightarrow \nabla^2 \psi$

Example No 1:

Convert the polar coordinates $(3, \frac{\pi}{2})$ as rectangular coordinates.

$$x = r\cos\theta$$

$$= 3\cos \pi/2$$

$$= 3(0) = 0$$

$$\boxed{x = 0}$$

$$y = r\sin\theta$$

$$= 3\sin \pi/2$$

$$= 3(1) = 3$$

$$\boxed{y = 3}$$

The rectangular coordinates are (0,3).

Imp

Example No 2:

Convert the rectangular coordinates (3,3) to polar coordinate.

$$r^2 = x^2 + y^2$$

$$r^2 = (3)^2 + (3)^2 \qquad \Rightarrow r^2 = 9 + 9$$

$$r = \sqrt{18} \qquad \Rightarrow \boxed{r = 3\sqrt{2}}$$

$$\text{Tan}\,\theta = \frac{y}{x}$$

$$\text{Tan}\,\theta = \frac{3}{3} \qquad \Rightarrow \text{Tan}\,\theta = 1$$

$$\theta = \text{Tan}^{-1}(1) \qquad \Rightarrow \theta = \frac{\pi}{4}$$

Hence the polar coordinates are $(3\sqrt{2}2, \frac{\pi}{4})$.

1071 1061

$$dS^2 = dr^2 + r^2 d\theta^2$$

Example. Convert the cylindrical coordinates $(4, \pi, 1)$ in rectangular coordinates.

Solution: Using the transformation:

$x = r\cos\theta = 4\cos\pi = -4$

$y = r\sin\theta = 4\sin\pi = 0$

$z = z = 1$

$\Rightarrow (4, \pi, 1) \longrightarrow (-4, 0, 1)$

Example: Convert $(1, 1, -1)$ in cylindrical coordinates.

using the inverse transformation:

$r = \sqrt{x^2+y^2} = \sqrt{1+1} = \sqrt{2}$

$\theta = \tan^{-1}\frac{y}{x} = \tan^{-1}(1) = \pi/4$

$z = -1$

$(1, 1, -1) \longrightarrow (\sqrt{2}, \pi/4, -1)$

Scale factors:

The scale factor for the cylindrical coordinates are given by

$$h_1 = h_r = \left|\frac{\partial \bar{r}}{\partial r}\right| = \sqrt{\cos^2\theta + \sin^2\theta} = 1$$

$$h_2 = h_\theta = \left|\frac{\partial \bar{r}}{\partial \theta}\right| = \sqrt{(-r\sin\theta)^2 + (r\cos\theta)^2}$$

$$= r\sqrt{\cos^2\theta + \sin^2\theta} = r$$

$$h_3 = h_z = \left|\frac{\partial \bar{r}}{\partial z}\right| = |k| = 1$$

hence the scale factor are:

$$\boxed{h_1 = h_r = 1} \qquad \boxed{h_2 = h_\theta = r} \qquad \boxed{h_3 = h_z = 1}$$

Example: Express the following in Spherical coordinates.

a) the Sphere $x^2+y^2+z^2=9$

using transformation equations:

$$(\rho\sin\theta\cos\phi)^2+(\rho\sin\theta\sin\phi)^2+(\rho\cos\theta)^2=9$$

$$\rho^2\sin^2\theta(\cos^2\phi+\sin^2\phi)+\rho^2\cos^2\theta=9$$

$$\Rightarrow \rho^2(\sin^2\theta+\cos^2\theta)=9$$

$$\Rightarrow \rho^2=9 \quad \text{or} \quad \rho=3$$

e) the plane $z=0$

$$\Rightarrow \rho\cos\theta=0$$

$$\rho\neq 0 \Rightarrow \cos\theta=0$$

$$\Rightarrow \theta=\cos^{-1}(0)$$

$$\Rightarrow \theta=\pi/2$$

b) the paraboloid: $z = x^2 + y^2$

$\Rightarrow \rho\cos\theta = \rho^2\sin^2\theta\cos^2\phi + \rho^2\sin^2\theta\sin^2\phi$

$\rho\cos\theta = \rho^2\sin^2\theta$

$\Rightarrow \cos\theta = \rho\sin^2\theta$

c) the plane $y = x$

$\Rightarrow \rho\sin\theta\sin\phi = \rho\sin\theta\cos\phi$

$\Rightarrow \tan\phi = 1 \Rightarrow \phi = \tan^{-1}(1)$

$\Rightarrow \phi = \pi/4$ and $5\pi/4$ b/c $0 \le \phi \le 2\pi$

d) the cone $z^2 = 3(x^2+y^2)$

$\Rightarrow \rho^2\cos^2\theta = 3\rho^2\sin^2\theta(\cos^2\varphi + \sin^2\varphi)$.

$\Rightarrow \tan^2\theta = 1/3$

$\Rightarrow \tan\theta = 1/\sqrt{3}$

$\Rightarrow \theta = \tan^{-1}(1/\sqrt{3})$

$\Rightarrow \theta = \pi/6$ b/c $0 \le \theta \le \pi$

Regard to Ali G
Remember in your precious prayers

Contravariant vectors:

- A contravariant vector or tangent vector is often abbreviated simply as vector, such as a direction vector or velocity vector.
- The contravariant vector has components that contra-vary with a change of basis to compensate.
- That is, the matrix that transforms the vector components must be the inverse of the matrix that transforms the basis vectors.
- Examples of vectors with contravariant components include the position of an object relative to an observer, or any derivative of position with respect to time, including velocity and acceleration.
- In Einstein notation, contravariant components are denoted with upper indices.

Covariant Vectors:

- In contrast, co-vectors also called dual vectors typically have units of the inverse of distance.
- An example of a co-vector is the gradient, which has units of a spatial derivative, or distance inverse. The components of co-vectors change in the same way as changes to scale of the reference axes and consequently are called covariant.
- A covariant vector often abbreviated as co-vector has components that **co-vary** with a change of basis.
- That is, the components must be transformed by the same matrix as the change of basis matrix.
- In Einstein notation, covariant components are denoted with lower indices.

Transformation Law for Contravariant tensor:

A vector field $\vec{A}$ is a contravariant tensor of order one provided its components A^i and $\tilde{A}^i$ relative to the coordinate systems x^i and $\tilde{x}^i$ obey the transformation

$$\tilde{A}^i = A^r \frac{\partial \tilde{x}^i}{\partial x^r} \quad ; 1 \leq i \leq n$$

$$A_i = a_{ij} \tilde{A}_j$$

Transformation Law for Covariant tensor:

A vector field $\vec{A}$ is a covariant tensor of order one provided its components A_i and $\tilde{A}_i$ relative to the coordinate systems x^i and $\tilde{x}^i$ obey the transformation

$$\tilde{A}_i = A_r \frac{\partial x^r}{\partial \tilde{x}^i} \quad ; 1 \leq i \leq n$$

Contravariant tensor:

The matrix field V is a contravariant tensor of order two if its components T^{ij} in x^i-system and $\tilde{T}^{ij}$ in $\tilde{x}^i$-system obey the law of transformation

$$\tilde{T}^{ij} = T^{rs} \frac{\partial \tilde{x}^i}{\partial x^r} \frac{\partial \tilde{x}^j}{\partial x^s}$$

Covariant tensor:

The matrix field V is a covariant tensor of order two if its components T_{ij} in x^i-system and $\tilde{T}_{ij}$ in $\tilde{x}^i$-system obey the law of transformation

$$\tilde{T}_{ij} = T_{rs} \frac{\partial x^r}{\partial \tilde{x}^i} \frac{\partial x^s}{\partial \tilde{x}^j}$$

Mixed tensor:

The matrix field V is a mixed tensor of order two, contravariant of order one and covariant of order one, if its components T_j^i in x^i-system and $\tilde{T}_{ij}^{i}$ in $\tilde{x}^i$-system obey the law of transformation

$$\tilde{T}_j^i = T_s^r \frac{\partial \tilde{x}^i}{\partial x^r} \frac{\partial x^s}{\partial \tilde{x}^j}$$

$$\tilde{T}_{21} = \frac{2(x')}{-x^2} = \frac{2(x')^2}{-x^2}$$

$$\tilde{T}_{22} = \frac{1}{x'}$$

Question: T^{ij}_{klm} is a mixed tensor of order five. Prove that T^{ij}_{kij} is a covariant tensor of order one.

Solution:

$$\tilde{T}^{ij}_{klm} = T^{rs}_{tuv} \frac{\partial \tilde{x}^i}{\partial x^r} \frac{\partial \tilde{x}^j}{\partial x^s} \frac{\partial x^t}{\partial \tilde{x}^k} \frac{\partial x^u}{\partial \tilde{x}^l} \frac{\partial x^v}{\partial \tilde{x}^m}$$

set $l = i$ and $m = j$:

$$\tilde{T}^{ij}_{kij} = T^{rs}_{tuv} \frac{\partial \tilde{x}^i}{\partial x^r} \frac{\partial \tilde{x}^j}{\partial x^s} \frac{\partial x^t}{\partial \tilde{x}^k} \frac{\partial x^u}{\partial \tilde{x}^i} \frac{\partial x^v}{\partial \tilde{x}^j}$$

$$= T^{rs}_{tuv} \left(\frac{\partial \tilde{x}^i}{\partial x^r} \frac{\partial x^u}{\partial \tilde{x}^i}\right)\left(\frac{\partial \tilde{x}^j}{\partial x^s} \frac{\partial x^v}{\partial \tilde{x}^j}\right) \frac{\partial x^t}{\partial \tilde{x}^k}$$

$$\tilde{T}^{ij}_{kij} = T^{rs}_{tuv} \delta^u_r \delta^v_s \frac{\partial x^t}{\partial \tilde{x}^k}$$

$\tilde{T}_k \qquad T^{rs}_{trs} \to T_t \qquad \frac{\partial x^u}{\partial x^r} = \delta^u_r$

OR

$$\tilde{T}_k = T_t \frac{\partial x^t}{\partial \tilde{x}^k}$$

Alternative Formulas (For tensors of order 2)

In terms of Jacobian matrix:

where $J = \frac{\partial(\tilde{x}^1, \tilde{x}^2)}{\partial(x^1, x^2)}$, $J^{-1} = \frac{\partial(x^1, x^2)}{\partial(\tilde{x}^1, \tilde{x}^2)}$

The transformation law for contravariant tensor is: $\tilde{T} = J T J^t$

Similarly, for a covariant tensor:

$$\tilde{T} = (J^{-1})^t T J^{-1}$$

and for mixed tensors:

$$\tilde{T} = J T J^{-1}$$

Question: Compute the formulas for the tensor components $\bar{T}^{ij}$ in polar coordinates if the components T^{ij} in rectangular coordinates are given.

Solution: let $T = [T^{ij}]$

and $\bar{T} = J T J^t$

where $J = \frac{\partial(r,\theta)}{\partial(x,y)}$ and $J^{-1} = \frac{\partial(x,y)}{\partial(r,\theta)}$

Question: Show that δ^i_j is an invariant.

Solution:

$$\tilde{\delta}^i_j = \frac{\partial \tilde{x}^i}{\partial x^k}\left(\frac{\partial x^l}{\partial \tilde{x}^j}\,\delta^k_l\right)$$

$$= \frac{\partial \tilde{x}^i}{\partial x^k}\,\frac{\partial x^k}{\partial \tilde{x}^j}$$

$$= \frac{\partial \tilde{x}^i}{\partial \tilde{x}^j} = \delta^i_j$$

$\Rightarrow$ Kronecker delta has same components in every coordinate system.

Question: From the tensors $S = S^{ij}$ and $T = T_{klm}$ form the inner product $U^{ij}_{km} = S^{ij} T_{klm}$ and show that it is contravariant of order 1 and covariant of order 2.

Solution: Consider,

$$U^{ij}_{km} = S^{ij} T_{klm}$$

$$\Rightarrow \tilde{U}^{ij}_{km} = \tilde{S}^{ij} \tilde{T}_{klm}$$

Symmetric Tensors

A tensor is said to symmetric with respect to two contravariant or two covariant indices if its components remain unchanged on an interchange of the two indices.

For example: (1) The tensor A^{ij} is Symmetric if $\boxed{A^{ij} = A^{ji}}$

(2) The tensor A^{ijk}_{lm} is symmetric w.r.t indices i and j if: $A^{ijk}_{lm} = A^{jik}_{lm}$

and w.r.t indices l and m if:

$$\boxed{A^{ijk}_{lm} = A^{ijk}_{ml}}$$

Skew symmetric tensor

A tensor is said to be skew symmetric with respect to two contravariant or two covariant indices if its components change sign on interchange of the two indices.

For example: (1) The tensor A^{ij} is skew-Symmetric if $\boxed{A^{ij} = -A^{ji}}$

(2) The tensor A^{ijk}_{lm} is skew-Symmetric w.r.t the indices j and k if:

$$\boxed{A^{ijk}_{lm} = -A^{ikj}_{lm}}$$

Conjugate symmetric tensor:

Consider a covariant symmetric tensor A_{ij} of rank two. Let d denote the determinant of this tensor i.e., $d = |A_{ij}|$ and $d \neq 0$. Then define A^{ij} by

$$A^{ij} = \frac{\text{Cofactor of } A_{ij} \text{ in the determinant } |A_{ij}|}{d}$$

'A^{ij}' is a contravariant symmetric tensor of rank two which is called conjugate OR reciprocal tensor of A_{ij}.

Metric Tensor:

This is a rank two tensor and is also called the fundamental tensor.

The main purpose of the metric tensor is to generalize the concept of distance to general curvilinear coordinate frames and maintain the invariance of distance in different coordinate systems.

Distance between 2 points:

Consider a rubber sheet

Important Result:

If the Jacobian matrix of the transformation from a given coordinate system x^i to a rectangular coordinate system $\bar{x}^i$ is

$$J = \frac{\partial \bar{x}^i}{\partial x^j},$$

then the matrix $g = [g_{ij}]$ of the Euclidean metric tensor in the x^i-System is given by

$$g = J^t J$$

Metric Tensor and Arc Lengths:

Arc length of a curve $x^i = x^i(t)$ is generally defined as:

$$L = \int_a^b \left| \frac{d\vec{r}}{dt} \right| dt \qquad ds^2 = g_{ij}\, dx^i\, dx^j$$

Now $\vec{r} = x^1 \hat{e}_1 + x^2 \hat{e}_2$

$$\left|\frac{d\vec{r}}{dt}\right| = \sqrt{g_{ij}\frac{dx^i}{dt}\frac{dx^j}{dt}}$$

$$\Rightarrow \boxed{L = \int_a^b \sqrt{g_{ij}\frac{dx^i}{dt}\frac{dx^j}{dt}}\, dt}$$

Conjugate Metric Tensor:

The inverse of the fundamental tensor or metric tensor

$$\left[g^{ij}\right]_{nn} = \left[g_{ij}\right]^{-1}_{nn}$$

is called the conjugate metric tensor.

* $g_{ij} = \hat{e}_i \cdot \hat{e}_j$

then $g^{ij} = \hat{e}'_i \cdot \hat{e}'_j$

where $\hat{e}_i$ and $\hat{e}'_i$ are the two sets of basis vectors associated normal to the coordinate surfaces and tangent to the coordinate curves.

The covariant and contravariant metric tensors are inverses of each other that is

$$[g_{ij}] = [g^{ij}]^{-1}$$

and

$$[g^{ij}] = [g_{ij}]^{-1}$$

$$\Rightarrow g_{ik} g^{kj} = \delta_i{}^j$$

and $$g^{ik} g_{kj} = \delta^i{}_j$$

It is common to call covariant form as metric tensor and contravariant form as inverse metric, associate metric, conjugate metric or reciprocal metric tensor.

$$g_{ij} = \begin{bmatrix} \hat{e}_1 \cdot \hat{e}_1 & \hat{e}_1 \cdot \hat{e}_2 & \cdots & \hat{e}_1 \cdot \hat{e}_n \\ & \hat{e}_2 \cdot \hat{e}_2 & \cdots & \\ \hat{e}_n \cdot \hat{e}_1 & & & \hat{e}_n \cdot \hat{e}_n \end{bmatrix}$$

For orthogonal coordinate system the metric tensor is diagonal, i.e.,

$$\boxed{g_{ij} = g^{ij} = 0} \quad \text{for } i \neq j$$

Question: Given that V^i is a contravariant vector $\Rightarrow$

Concept of Length and Angle:

The norm (or length) of an arbitrary vector V^2 or V_2 is the non-negative real number.

$$\|\vec{V}\| = \sqrt{V^2} = \sqrt{V_2 V^2}$$

Concept of Length and Angle:

The norm (or length) of an arbitrary vector V^i or V_i is the non-negative real number.

$$\|\vec{V}\| = \sqrt{V^2} = \sqrt{V_i V^i}$$

The angle θ between two non-null contravariant vectors $\vec{U}$ and $\vec{V}$ is defined by:

$$\cos\theta = \frac{\vec{U}\cdot\vec{V}}{\|\vec{U}\|\,\|\vec{V}\|} = \frac{g_{ij}\,U^i V^j}{\sqrt{g_{pq}U^p U^q}\,\sqrt{g_{rs}V^r V^s}}$$

3) $\nabla(\vec{a}.\vec{r}) = \vec{a}$, where $\vec{a}$ is any constant vector.

Proof: consider L.H.S, where

$\vec{a}.\vec{r} = a_j x_j$ and $[\nabla\phi]_i = \partial_i \phi$

$$\Rightarrow \left[\nabla(\vec{a}.\vec{r})\right]_i = \partial_i(a_j x_j)$$

$$= a_j \partial_i x_j + x_j \underbrace{\partial_i a_j}_{\to 0} \quad \text{(product rule)}$$

$$= a_j \partial_i x_j$$

$$= a_j \delta_{ij}$$

$$= a_i$$

$$\Rightarrow \nabla(\vec{a}.\vec{r}) = \vec{a}$$

2) $\nabla \times \vec{r} = \vec{0}$

Proof:

$$[\nabla \times \vec{r}]_i = \varepsilon_{ijk}\, \partial_j x_k \quad \text{(definition of curl)}$$

$$= \varepsilon_{ijk}\, \delta_{kj} \quad \left(\because \partial_j x_k = \frac{\partial x_k}{\partial x_j}\right)$$

$$= \varepsilon_{ijj} \quad \text{(using def. of } \delta_{ij})$$

$$= 0 \quad \text{(by the def. of } \varepsilon_{ijk})$$

4) $\nabla \cdot \nabla\phi = \nabla^2\phi$ $\qquad \nabla \cdot \vec{A} = \partial_i A_i$, with $\vec{A} \to \nabla\phi$

Proof: consider,

$$\nabla \cdot \nabla\phi = \partial_i [\nabla\phi]_i \quad \left(\because \nabla_i = \partial_i,\ [\nabla\phi]_i = \partial_i\phi\right)$$

$$= \partial_i (\partial_i \phi)$$

$$= \partial_i \partial_i \phi$$

$$= \partial_{ii}\phi$$

$$= \nabla^2\phi$$

4) $\nabla \cdot \nabla\phi = \nabla^2\phi$

$\nabla \cdot \vec{A} = \partial_i A_i$

$\vec{A} \to \nabla\phi$

Proof: consider,

$$\nabla \cdot \nabla\phi = \partial_i [\nabla\phi]_i$$

$$= \partial_i (\partial_i \phi)$$

$$= \partial_i \partial_i \phi$$

$$= \partial_{ii} \phi$$

$$= \nabla^2 \phi$$

$(\because \nabla_i = \partial_i$

$[\nabla\phi]_i = \partial_i \phi)$

Regard to Ali G

Remember in your precious prayers

Question: Find metric tensor for cylindrical coordinates x^i, connected to rectangular coordinates $\bar{x}^i$ through the relations:

$$\bar{x}^1 = x^1 \cos x^2, \quad \bar{x}^2 = x^1 \sin x^2, \quad \bar{x}^3 = x^3$$

Solution: First we find Jacobian matrix:

$$J = \left[\frac{\partial \bar{x}^i}{\partial x^j}\right] = \begin{bmatrix} \cos x^2 & -x^1 \sin x^2 & 0 \\ \sin x^2 & x^1 \cos x^2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Hence the Euclidean metric for cylindrical coordinates is given by:

$$g = J^t J = \begin{bmatrix} \cos x^2 & \sin x^2 & 0 \\ -x^1 \sin x^2 & x^1 \cos x^2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos x^2 & -x^1 \sin x^2 & 0 \\ \sin x^2 & x^1 \cos x^2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & (x^1)^2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Question: Find the components g_{ij} of the Euclidean metric tensor in the special coordinate system x^i, related to rectangular coordinates $\bar{x}^i$ by:

$$x^1 = \bar{x}^1, \quad x^2 = e^{\bar{x}^2 - \bar{x}^1}$$

$$J = \begin{bmatrix} \frac{\partial \bar{x}^1}{\partial x^1} & \frac{\partial \bar{x}^1}{\partial x^2} \\ \frac{\partial \bar{x}^2}{\partial x^1} & \frac{\partial \bar{x}^2}{\partial x^2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1/x^2 \end{bmatrix}$$

$\star\star$ $\bar{x}^1 = x^1$ and $\bar{x}^2 = x^1 + \ln x^2$

Implies:

$$g = J^t J = \begin{bmatrix} 1 & 1 \\ 0 & 1/x^2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1/x^2 \end{bmatrix}$$

$$= \begin{bmatrix} 1+1 & 1/x^2 \\ 1/x^2 & (1/x^2)^2 \end{bmatrix} = \begin{bmatrix} 2 & \frac{1}{x^2} \\ \frac{1}{x^2} & (\frac{1}{x^2})^2 \end{bmatrix}$$

Metric Tensor and Arc Lengths:

Arc length of a curve $x^i = x^i(t)$ is generally defined as:

$$L = \int_a^b \left| \frac{d\vec{r}}{dt} \right| dt \qquad ds^2 = g_{ij}\, dx^i dx^j$$

Now $\vec{r} = x^1 \hat{e}_1 + x^2 \hat{e}_2$

$$\frac{d\vec{r}}{dt} = \frac{\partial \vec{r}}{\partial x^1}\frac{dx^1}{dt} + \frac{\partial \vec{r}}{\partial x^2}\frac{dx^2}{dt}$$

$$\left|\frac{d\vec{r}}{dt}\right|^2 = \frac{d\vec{r}}{dt} \cdot \frac{d\vec{r}}{dt}$$

$$= \left(\frac{dx^1}{dt}\right)^2 \left(\frac{\partial \vec{r}}{\partial x^1} \cdot \frac{\partial \vec{r}}{\partial x^1}\right) + 2\frac{dx^1}{dt}\frac{dx^2}{dt}\left(\frac{\partial \vec{r}}{\partial x^1} \cdot \frac{\partial \vec{r}}{\partial x^2}\right)$$

$$+ \left(\frac{dx^2}{dt}\right)^2 \left(\frac{\partial \vec{r}}{\partial x^2} \cdot \frac{\partial \vec{r}}{\partial x^2}\right)$$

$$\Rightarrow \left|\frac{d\vec{r}}{dt}\right|^2 = \frac{dx^i}{dt}\frac{\partial \vec{r}}{\partial x^i} \cdot \frac{dx^j}{dt}\frac{\partial \vec{r}}{\partial x^j}$$

$$= \frac{dx^i}{dt}\frac{dx^j}{dt}\left(\frac{\partial \vec{r}}{\partial x^i} \cdot \frac{\partial \vec{r}}{\partial x^j}\right)$$

$$= \frac{dx^i}{dt}\frac{dx^j}{dt}(\hat{e}_i \cdot \hat{e}_j) \qquad (\because \hat{e}_i \cdot \hat{e}_j = g_{ij})$$

$$\left|\frac{d\vec{r}}{dt}\right| = \sqrt{g_{ij}\frac{dx^i}{dt}\frac{dx^j}{dt}}$$

$$\Rightarrow L = \int_a^b \sqrt{g_{ij}\frac{dx^i}{dt}\frac{dx^j}{dt}}\, dt$$

Example: Suppose that on $\mathbb{R}^3$ a matrix field is given in x^i-system by:

$$[g_{ij}] = \begin{bmatrix} (x^1)^2 - 1 & 1 & 0 \\ 1 & (x^2)^2 & 0 \\ 0 & 0 & 64/9 \end{bmatrix}$$

where $[(x^1)^2 - 1](x^2)^2 \neq 1$

For this metric, compute the arc-length parameter and length of the given curve.